Paper Review

Structural Causal Bandit : Where to Intervene?

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Preliminary : Multi-Armed Bandit

Introduction to MAB, and Regret

Multi-Armed Bandit (MAB)

Problem Definition

- Maximizing cumulated reward under uncertainty
  - $\mathcal{A} = \{a_1, a_2, \cdots, a_K\}$
  - Each Arm's Reward is independent to each other (original setting)

- Regret
  - Cumulative reward difference between optimal arm and selected arm
  - $R_T = T_{\mu^*} - \sum_{t=1} \mu_{a_t}$

T : total number of trials
μ^* : expected reward of the optimal arm
μ_{a_t} : expected reward of the arm chosen at time t

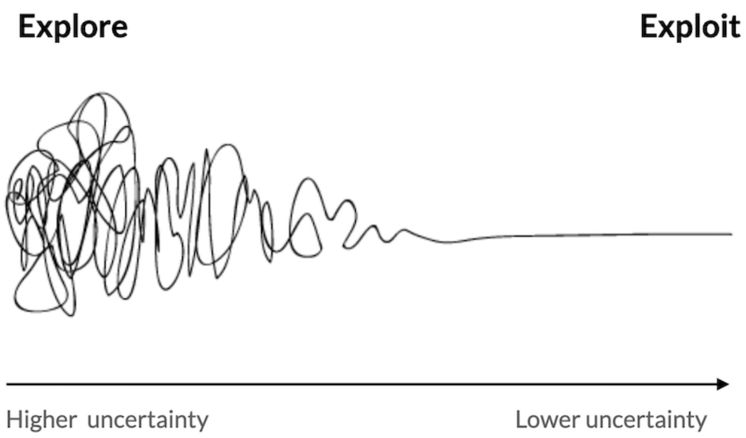
1) Reference : Lecture note of "Introduction to Reinforcement Learning", Jun-min So, Sogang University

Multi-Armed Bandit (MAB)

Exploration & Exploitation Trade-off

- We want to know the best arm
  - Many different strategies exist to find the best arm
  - Merely finding the best is not enough

- Trade-off
  - Exploration : Removing Uncertainty (Acquisition of Information)
  - Exploitation : Choosing the best arm we know (Greedy Selection)



Multi-Armed Bandit (MAB)

Regret Growth Rate

- ϵ-greedy
  - We select the "best" arm with probability 1 - ϵ, and a random arm with probability ϵ
  - Fixed ϵ : $\mathbb{E}[R_T] \approx \epsilon \cdot \delta \cdot T$

δ: Expected reward gap between the optimal and the suboptimal arm

Algorithm	Expected Regret	Main Idea
Random	$O(T)$	Pure exploration
ϵ-greedy	$O(\epsilon T)$, $O(\log T)$	Fixed exploration rate
UCB	$O(\log T)$	Optimistic exploration
Thompson Sampling	$O(\log T)$	Bayesian Posterior

Non-Trivial Dependencies Among Arms

Introduction to Linear Bandit, and SCM-MAB

Non-trivial dependencies among arms

Linear Bandit

- Reward Function
  - $Reward_i = \theta^T x_i + \epsilon$ (Shape Convention)
  - Now reward of arm gives you information for other arms

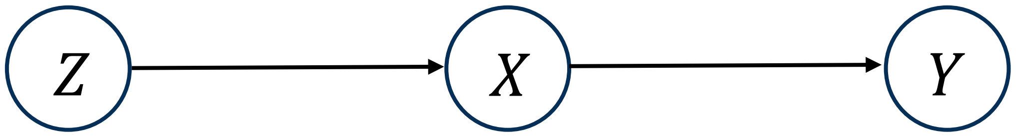
- Learning $\hat{\theta}$ = Learning Expected Value for Arms Simultaneously
  - LinUCB : UCB-variant for Linear Bandit
  - Regret for LinUCB, and UCB
  - LinUCB is better if there are lots of arms : $d < \sqrt{K}$

Algorithm	**Gap-dependent Regret**	**Gap-free Regret**
UCB	$O\left(\sum_{a:\Delta_a>0} \frac{\log T}{\Delta_a}\right)$	$O(\sqrt{KT\log T})$
LinUCB	$O\left(\frac{d\log T}{\Delta_{\min}}\right)$	$O(d\sqrt{T\log T})$

Non-trivial dependencies among arms

SCM-MAB : MAB with Structural Causal Model

- Action = Intervention
  - Pulling Arm in SCM-MAB is same with do-operation do(**X**=**x**)
  - Due to graph surgery, P(y|do(X), do(Z)) = P(y|do(X))
  - Skip trials for do(X=0, Z=0), do(X=0, Z=1) but do(X=0)



Possible Actions

$do(\emptyset)$

$do(X=0),\ do(X=1)$

$do(Z=0),\ do(Z=1)$

$do(X=0, Z=0),\ do(X=0, Z=1),\ do(X=1, Z=0),\ do(X=1, Z=1)$

SCM-MAB

Skipping Suboptimal Arms using Properties of Causal Diagram

Equivalence Among Arms

Recap : Do-Calculus

Theorem (Rules of Do-calculus Pearl [1995])

The following transformations are valid for any do-distribution induced by a causal model $\mathscr{M}$ *:*

Rule 1: *Adding/removing Observations*

$$P(\mathbf{y}|do(\mathbf{x}),\mathbf{z}) = P(\mathbf{y}|do(\mathbf{x})) \qquad \textit{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathscr{G}_{\overline{\mathbf{X}}}}.$$

Rule 2: *Action/observation Exchange*

$$P(\mathbf{y}|do(\mathbf{x}),do(\mathbf{z})) = P(\mathbf{y}|do(\mathbf{x}),\mathbf{z}) \qquad \textit{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathscr{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}.$$

Rule 3: *Adding/removing Actions*

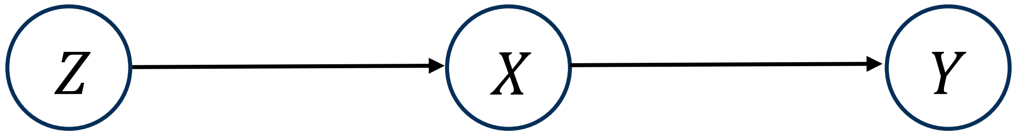
$$P(\mathbf{y}|do(\mathbf{x}),do(\mathbf{z})) = P(\mathbf{y}|do(\mathbf{x})) \qquad \textit{if } (\mathbf{Z} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{X})_{\mathscr{G}_{\overline{\mathbf{X}\mathbf{Z}}}}$$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- Skip Redundant Actions using **Rule 3**

Definition 1 (Minimal Intervention Set (MIS)). A set of variables $\mathbf{X} \subseteq \mathbf{V} \backslash \{Y\}$ is said to be a *minimal intervention set* relative to $[\![G, Y]\!]$ if there is no $\mathbf{X}' \subset \mathbf{X}$ such that $\mu_{\mathbf{x}[\mathbf{X}']} = \mu_{\mathbf{x}}$ for every SCM conforming to the G.



MIS in the Chain

$$do(X, Z) = do(X) \ \rightarrow \ \{X\}$$

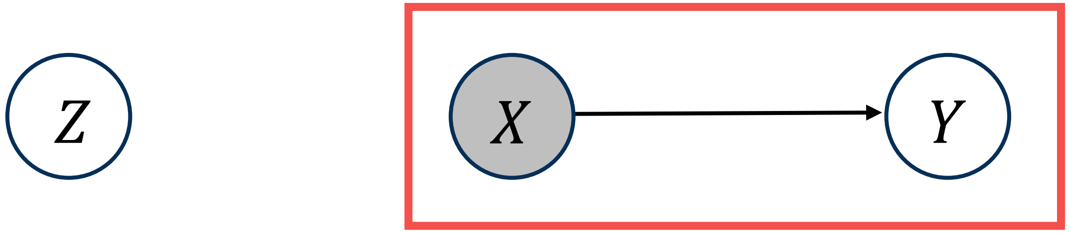
$$\therefore \ \text{MIS} : \{\emptyset, \ \{X\}, \ \{Z\}\}$$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- Skip Redundant Actions using **Rule 3**

Proposition 1 (Minimality). *A set of variables* $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ *is a minimal intervention set for* G *with respect to* Y *if and only if* $\mathbf{X} \subseteq an(Y)_{G_{\overline{\mathbf{X}}}}$.



MIS in the Chain

$do(X, Z) = do(X) \;\rightarrow\; \{X\}$

$\therefore\;$ MIS : $\{\emptyset,\; \{X\},\; \{Z\}\}$

Equivalence Among Arms

Minimal Intervention Set (MIS)

- We can skip further with Partial-orders among arms

$$Z \rightarrow X \rightarrow Y$$

$$\max_{z \in D(Z)} \mu_z \leq \max_{x \in D(X)} \mu_x$$

$$\mu_z^* = \sum_x \mu_x \cdot P(x|do(z^*)) \leq \sum_x \mu_x^* P(x|do(z^*)) = \mu_x^*$$

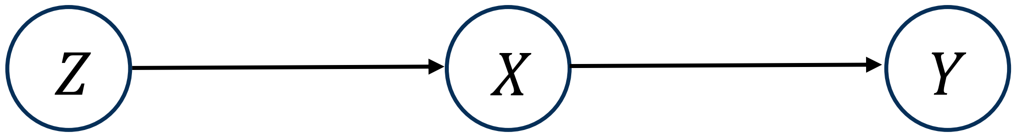
$\therefore$ do(X=0), do(X=1) : Possibly Optimal Interventions

Equivalence Among Arms

POMIS (Possibly Optimal Minimal Intervention Set)

- Our Goal : Find every POMIS in the given causal diagram

Definition 2 (Possibly-Optimal Minimal Intervention Set (POMIS))**.** Given information $[\![G, Y]\!]$, let $\mathbf{X}$ be a MIS. If there exists a SCM conforming to G such that $\mu_{\mathbf{x}^*} > \forall_{\mathbf{Z} \in \mathbb{Z} \setminus \{\mathbf{X}\}} \mu_{\mathbf{z}^*}$, where $\mathbb{Z}$ is the set of MISs with respect to G and Y, then $\mathbf{X}$ is a *possibly-optimal minimal intervention set* with respect to the information $[\![G, Y]\!]$.



We can easily confirm that Z, and empty set is not POMIS, but a X

$\therefore$ do(X=0), do(X=1) : Possibly Optimal Interventions

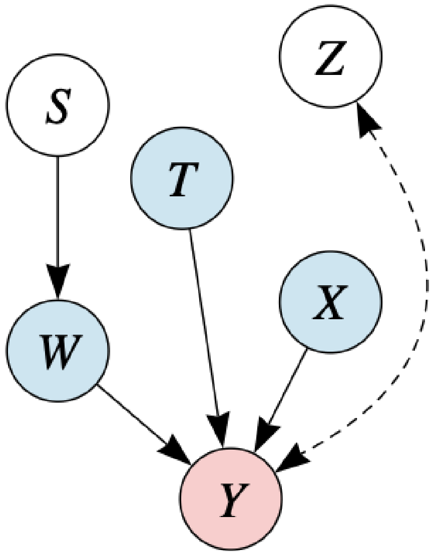
Equivalence Among Arms

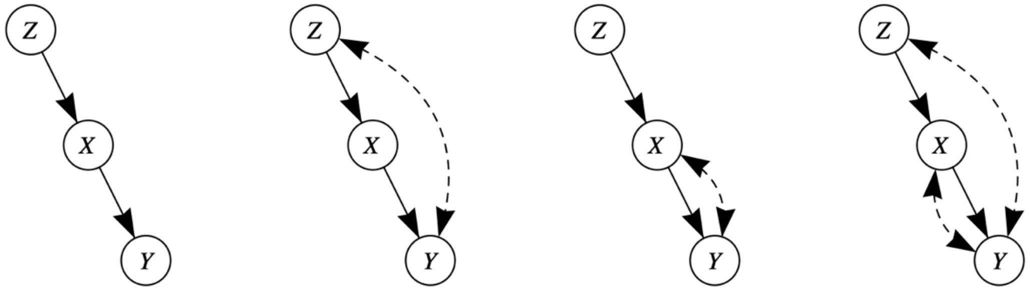
POMIS (Possibly Optimal Minimal Intervention Set)

- Our Goal : Find every POMIS in the given causal diagram

Proposition 2. *Given information* $[\![G, Y]\!]$, *if* Y *is not confounded with* $an(Y)_G$ *via unobserved confounders, then* $pa(Y)_G$ *is the only POMIS.*

Corollary 3 (Markovian POMIS). *Given* $[\![G, Y]\!]$, *if* G *is Markovian, then* $pa(Y)_G$ *is the only POMIS.*





(a) (b) (c) (d)

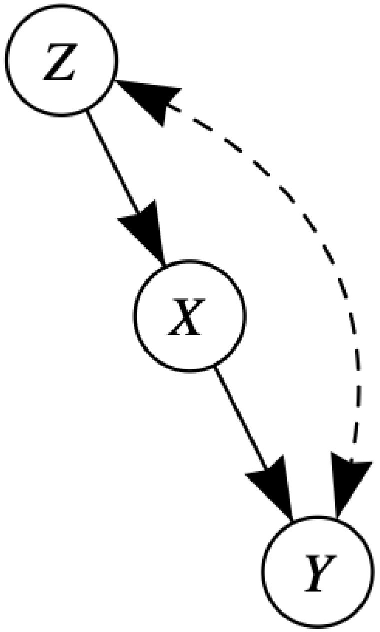
	$\emptyset$	$D(X)$	$D(Z)$
(a)		✓	
(b)	✓	✓	
(c)		✓	✓
(d)	✓	✓	✓

(e)

Equivalence Among Arms

POMISs of Graph with Unobserved Confounder

▸ Front-Door Case



Assume given variables are binary variables, and P(U=1) = P(U=0) = 0.5

$Z \leftarrow U$

$X \leftarrow Z$

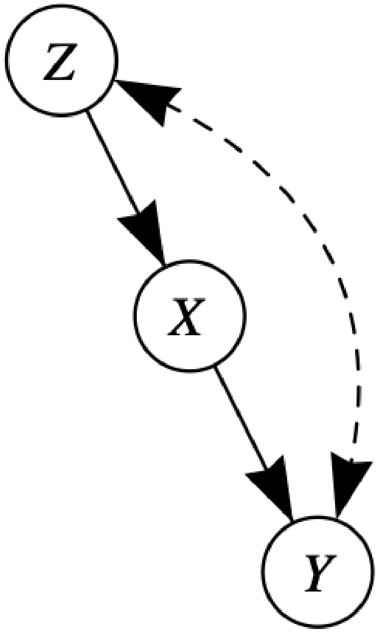
$Y \leftarrow \neg(X \oplus U)$

If we intervene Z or X, expected value of Y is 0.5
However, Y is always one if we do nothing

Equivalence Among Arms

POMISs of Graph with Unobserved Confounder

▸ Front-Door Case



$$\mathbb{E}[y \mid do(z^*)] = \sum_x \mu_x P(x \mid do(z^*)) \leq \sum_x \mu_x^* P(x \mid do(z^*)) = \mu_x^*$$

$\therefore$ POMISs $= \{\emptyset, \{X\}\}$

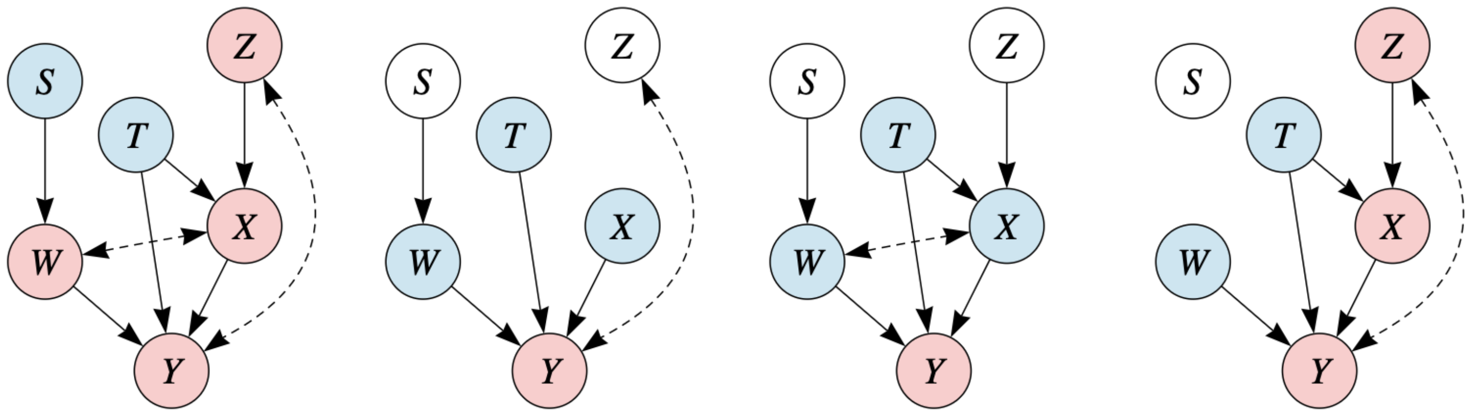
Q : How can we gain POMISs ?
A : Using Graphical Characterization of POMIS

Graphical Characterization of POMIS

MUCT (Minimal Unobserved-Confounders' Territory)

- Our Goal : Find every POMIS in the given causal diagram

Definition 3 (Unobserved-Confounders' Territory)**.** Given information $[\![G, Y]\!]$, let H be $G[An(Y)_G]$. A set of variables $\mathbf{T} \subseteq \mathbf{V}(H)$ containing Y is called an *UC-territory* on G with respect to Y if $De(\mathbf{T})_H = \mathbf{T}$ and $\mathrm{CC}(\mathbf{T})_H = \mathbf{T}$.



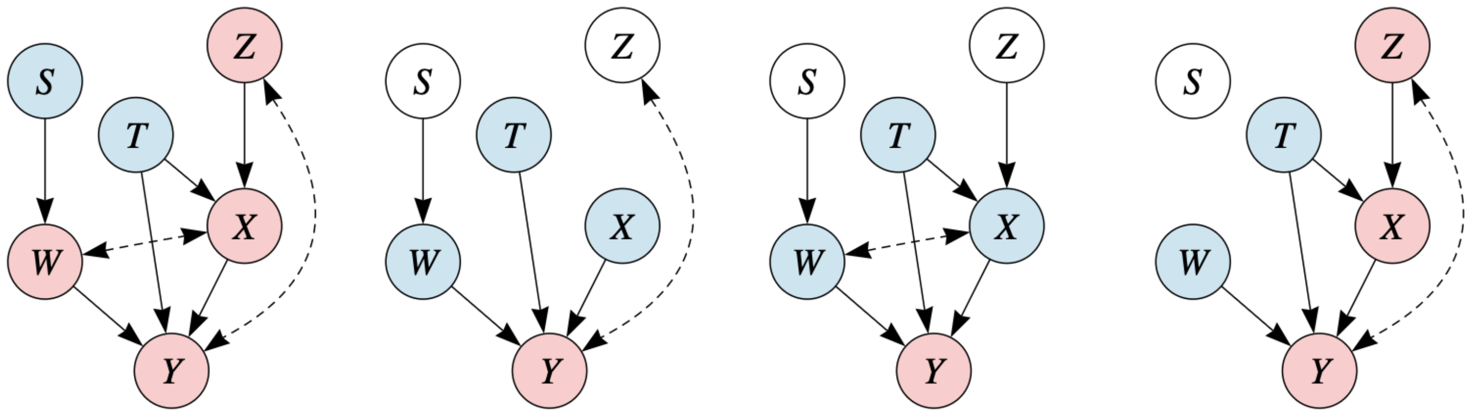
(a) G (b) $G_{\overline{X}}$ (c) $G_{\overline{Z}}$ (d) $G_{\overline{W}}$

Graphical Characterization of POMIS

MUCT (Minimal Unobserved-Confounders' Territory)

- Our Goal : Find every POMIS in the given causal diagram

Definition 4 (Interventional Border). Let $\mathbf{T}$ be a minimal UC-territory on G with respect to Y. Then, $\mathbf{X} = pa(\mathbf{T})_G \setminus \mathbf{T}$ is called an *interventional border* for G with respect to Y.



(a) G (b) $G_{\overline{X}}$ (c) $G_{\overline{Z}}$ (d) $G_{\overline{W}}$

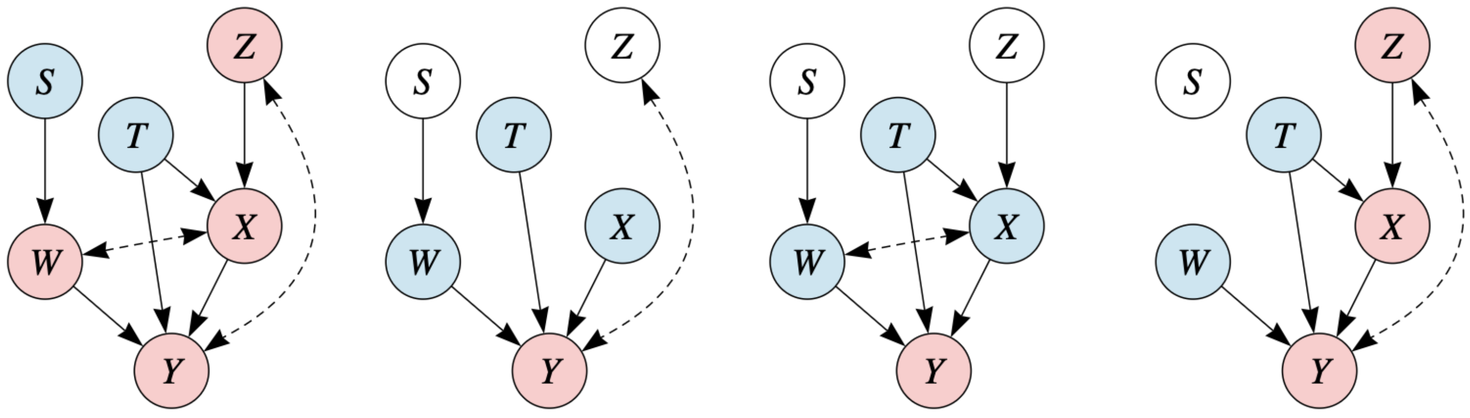
Graphical Characterization of POMIS

Intervention Border : Parents of MUCT

- Finding POMISs = Finding IBs

Proposition 4. $\mathsf{IB}(G, Y)$ *is a POMIS given* $[\![G, Y]\!]$.

Proposition 5. *Given* $[\![G, Y]\!]$, $\mathsf{IB}(G_{\overline{\mathbf{W}}}, Y)$ *is a POMIS, for any* $\mathbf{W} \subseteq \mathbf{V} \setminus \{Y\}$.



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Graphical Characterization of POMIS

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Theorem 6. *Given* $[\![G, Y]\!]$, $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ *is a POMIS if and only if* $\mathsf{IB}(G_{\overline{\mathbf{X}}}, Y) = \mathbf{X}$.

- Naive Approach
  - Enumerate every possible intervention set and get IB
  - **DFS-like** approach will do the job ! ! !

Algorithmic Characterization of POMIS

Properties for finding POMISs

- DFS-like POMIS Search

Proposition 7. *Let* $\mathbf{T}$ *and* $\mathbf{X}$ *be the* $\mathsf{MUCT}(G_{\overline{\mathbf{W}}}, Y)$ *and* $\mathsf{IB}(G_{\overline{\mathbf{W}}}, Y)$, *respectively, relative to* G *and* Y. *Then, for any* $\mathbf{Z} \subseteq \mathbf{V} \backslash \mathbf{T}$, $\mathsf{MUCT}(G_{\overline{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{T}$ *and* $\mathsf{IB}(G_{\overline{\mathbf{X} \cup \mathbf{Z}}}, Y) = \mathbf{X}$.

Proposition 8. *Let* $H = G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$ *where* $\mathbf{T}$ *and* $\mathbf{X}$ *are MUCT and IB given* $[\![G_{\overline{\mathbf{W}}}, Y]\!]$, *respectively. Then, for any* $\mathbf{W}' \subseteq \mathbf{T} \backslash \{Y\}$, $H_{\overline{\mathbf{W}'}}$ *and* $G_{\overline{\mathbf{W} \cup \mathbf{W}'}}$ *yield the same MUCT and IB with respect to* Y.

- We don't have to see outside of IB, and MUCT
  - Intervening outside variable does not affect Y (Rule 3 of Do-Calculus)
  - We don't have to see whole graph but a subgraph

Algorithmic Characterization of POMIS

Properties for finding POMISs

▸ DFS-like POMIS Search

Algorithm 1 Algorithm enumerating all POMISs with $[\![G, Y]\!]$

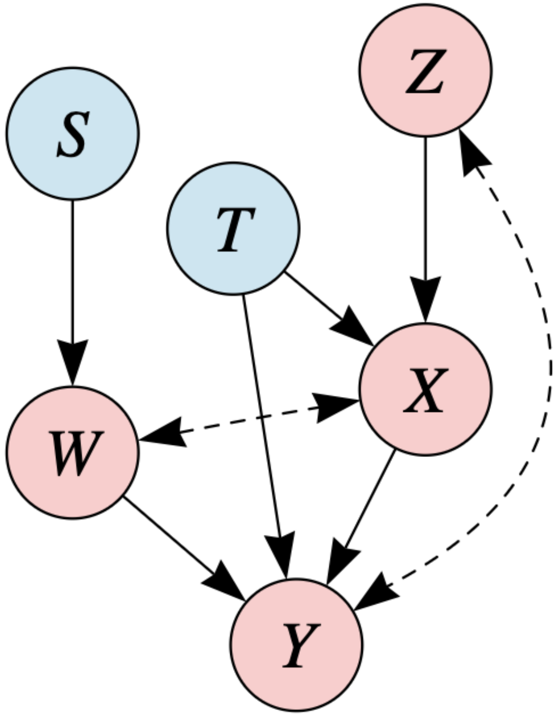
function POMISs(G, Y)
    $\mathbf{T}, \mathbf{X} = \text{MUCT}(G, Y), \text{IB}(G, Y); H = G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}]$
    return $\{\mathbf{X}\} \cup \text{subPOMISs}(H,\ Y,\ \text{reversed}(\text{topological-sort}(H)) \cap (\mathbf{T} \setminus \{Y\}),\ \emptyset)$

function subPOMISs(G, Y, $\boldsymbol{\pi}$, $\mathbf{O}$)
    $\mathbf{P} = \emptyset$
    for $\pi_i \in \boldsymbol{\pi}$ **do**
        $\mathbf{T},\ \mathbf{X},\ \boldsymbol{\pi}',\ \mathbf{O}' = \text{MUCT}(G_{\overline{\pi_i}}, Y),\ \text{IB}(G_{\overline{\pi_i}}, Y),\ \boldsymbol{\pi}^{i+1:|\boldsymbol{\pi}|} \cap \mathbf{T},\ \mathbf{O} \cup \boldsymbol{\pi}^{1:i-1}$
        if $\mathbf{X} \cap \mathbf{O}' = \emptyset$ **then**
            $\mathbf{P} = \mathbf{P} \cup \{\mathbf{X}\} \cup (\text{subPOMISs}(G_{\overline{\mathbf{X}}}[\mathbf{T} \cup \mathbf{X}],\ Y,\ \boldsymbol{\pi}',\ \mathbf{O}')$ **if** $\boldsymbol{\pi}' \neq \emptyset$ **else** $\emptyset)$
    return $\mathbf{P}$

Algorithmic Characterization of POMIS

Properties for finding POMISs

- **DFS-like POMIS Search**



Reversed topological order : W < X < Z

{W} : do(S,T,W) – IB = {W, T}
{W, X} : do(S,T,W,X) – IB = {W, T, X}
{W, X, Z} : do(S,T,W,X,Z) – IB = {W, T, X}

{X} : do(S,T,X) – IB = {W, T, X}
{X, Z} is pruned (graph containing do(W) is exhaustively used

{Z} : do(S,T,Z) – IB = {W, T, X}

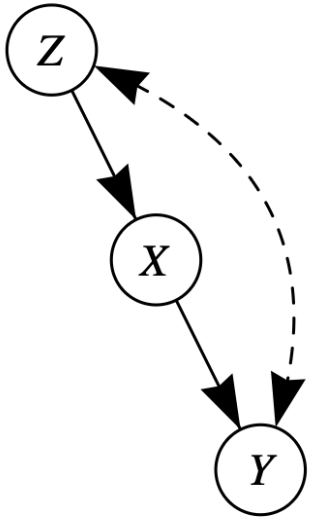
Experiments

Regret Comparison : UCB, KL-UCB, TS (with and without POMIS)

Experiments with Front-Door, Back-Door

Environment

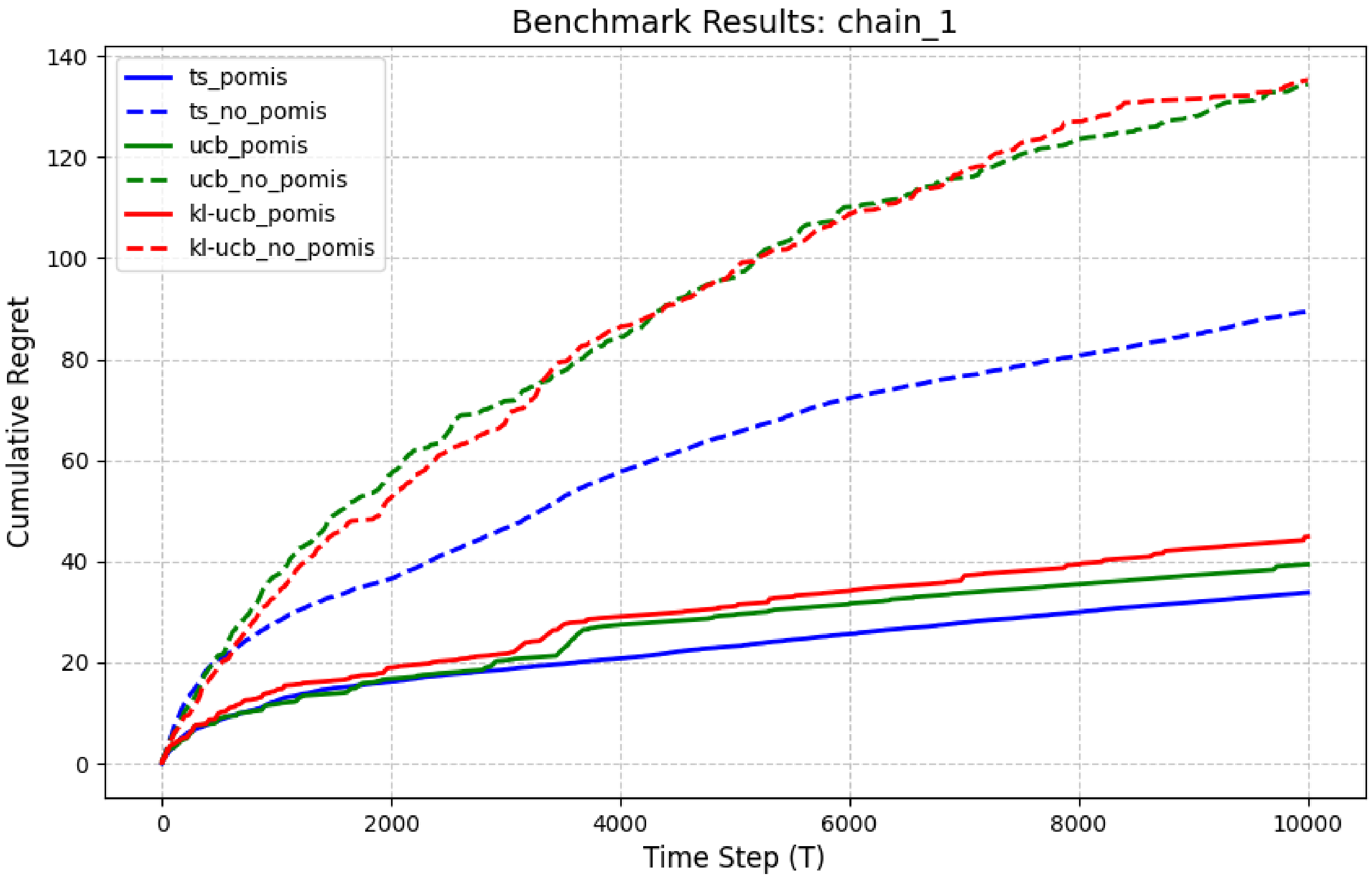
- **Variables in the Graph are binary (0-1)**
  - Bernoulli Sampling for determining the value of node
  - Weight is allocated for each edge just like Linear SCM
  - Value of node is sampled from distribution where P(V=1) = $\frac{1}{1+e^{-w^T \cdot Pa(V)}}$
  - 5 different random seeds were adopted

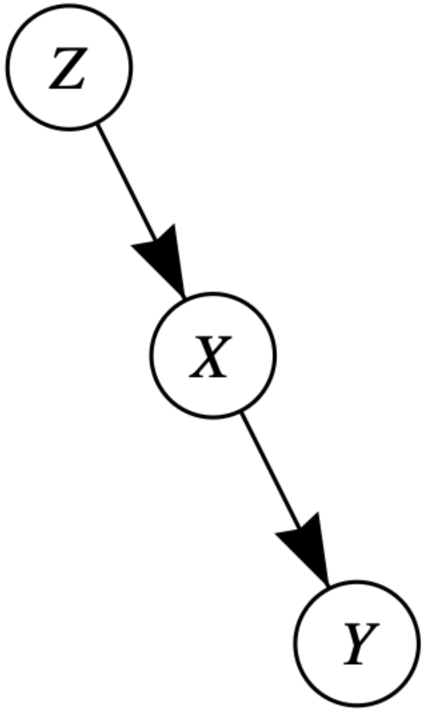


- We cannot know expected reward of each arm
  - Use Monte Carlo Approximation (or sample-based estimation)
  - We deem these estimated as oracle when we calculate the regret
  - Arm we have to try is significantly reduced : $9 \rightarrow 3$

Experiments with UCB, KL-UCB, TS

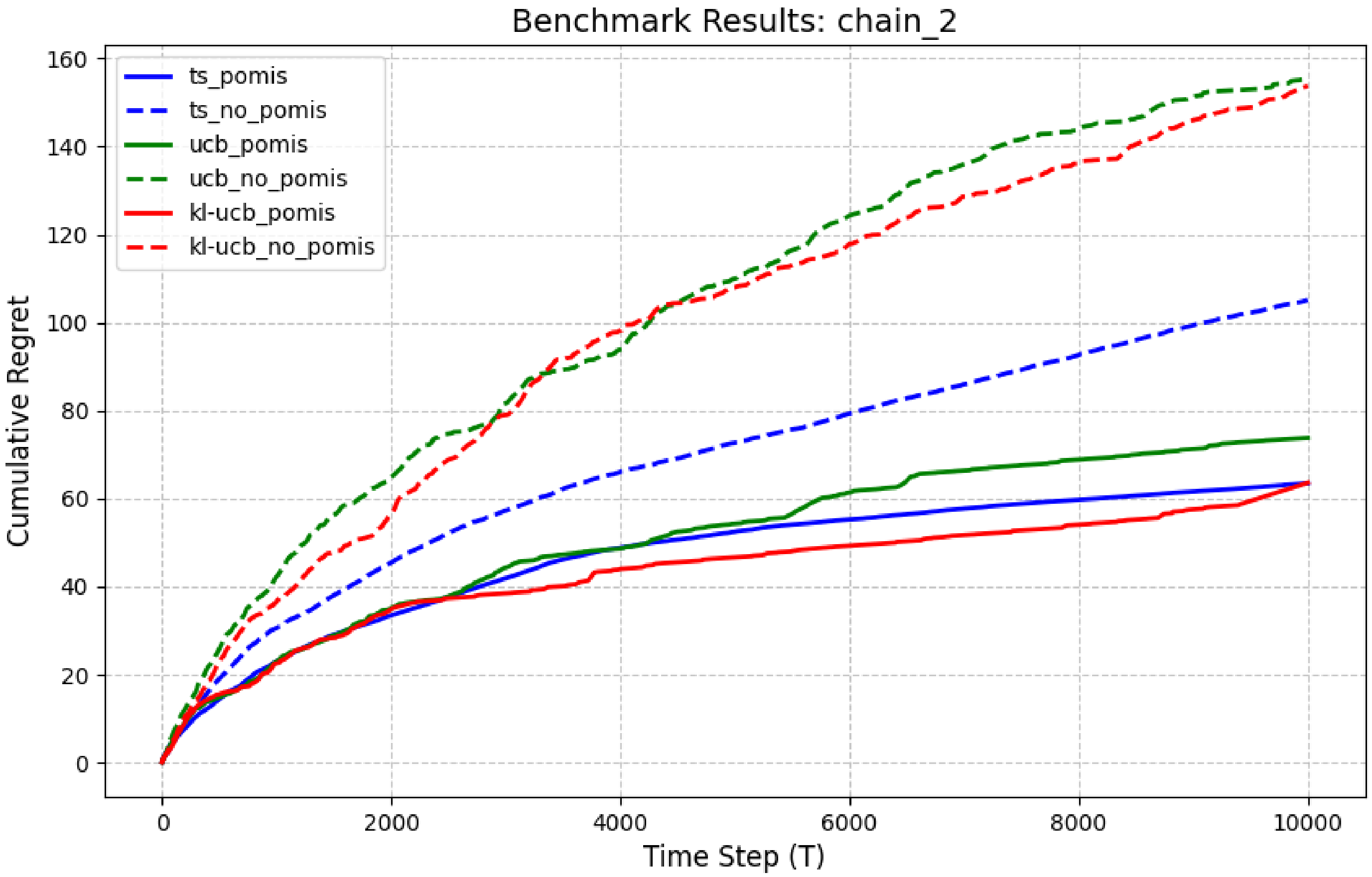
Markovian Model

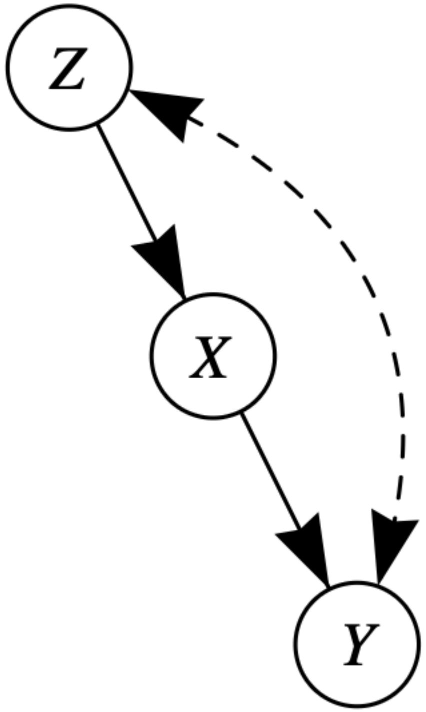




Experiments with UCB, KL-UCB, TS

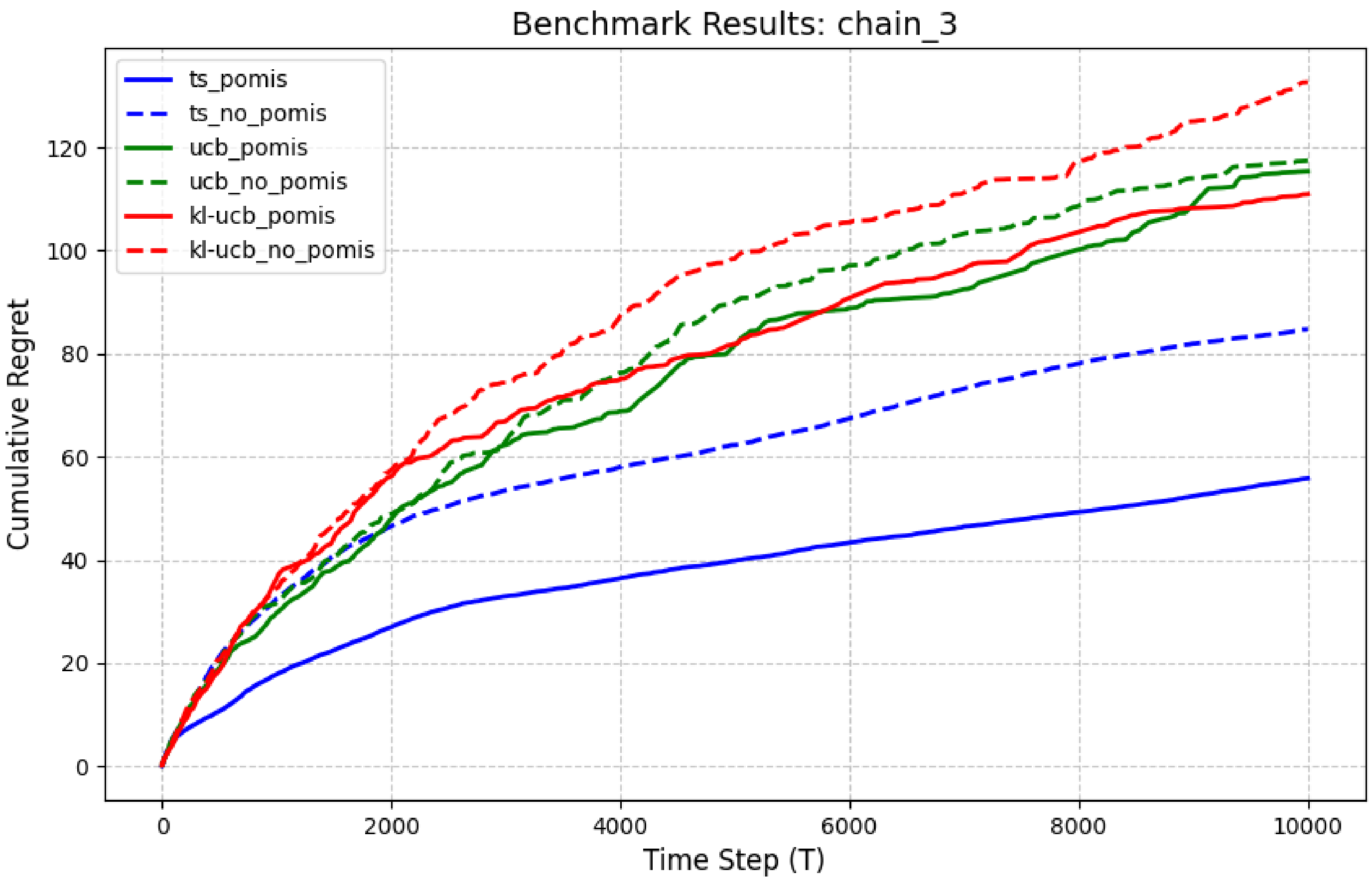
Front-Door Case

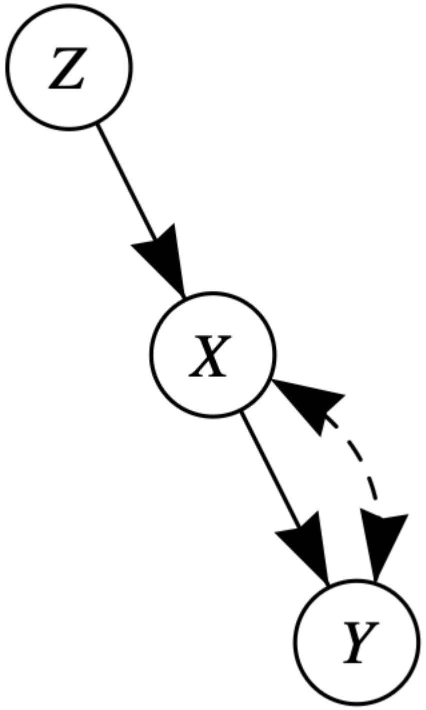




Experiments with UCB, KL-UCB, TS

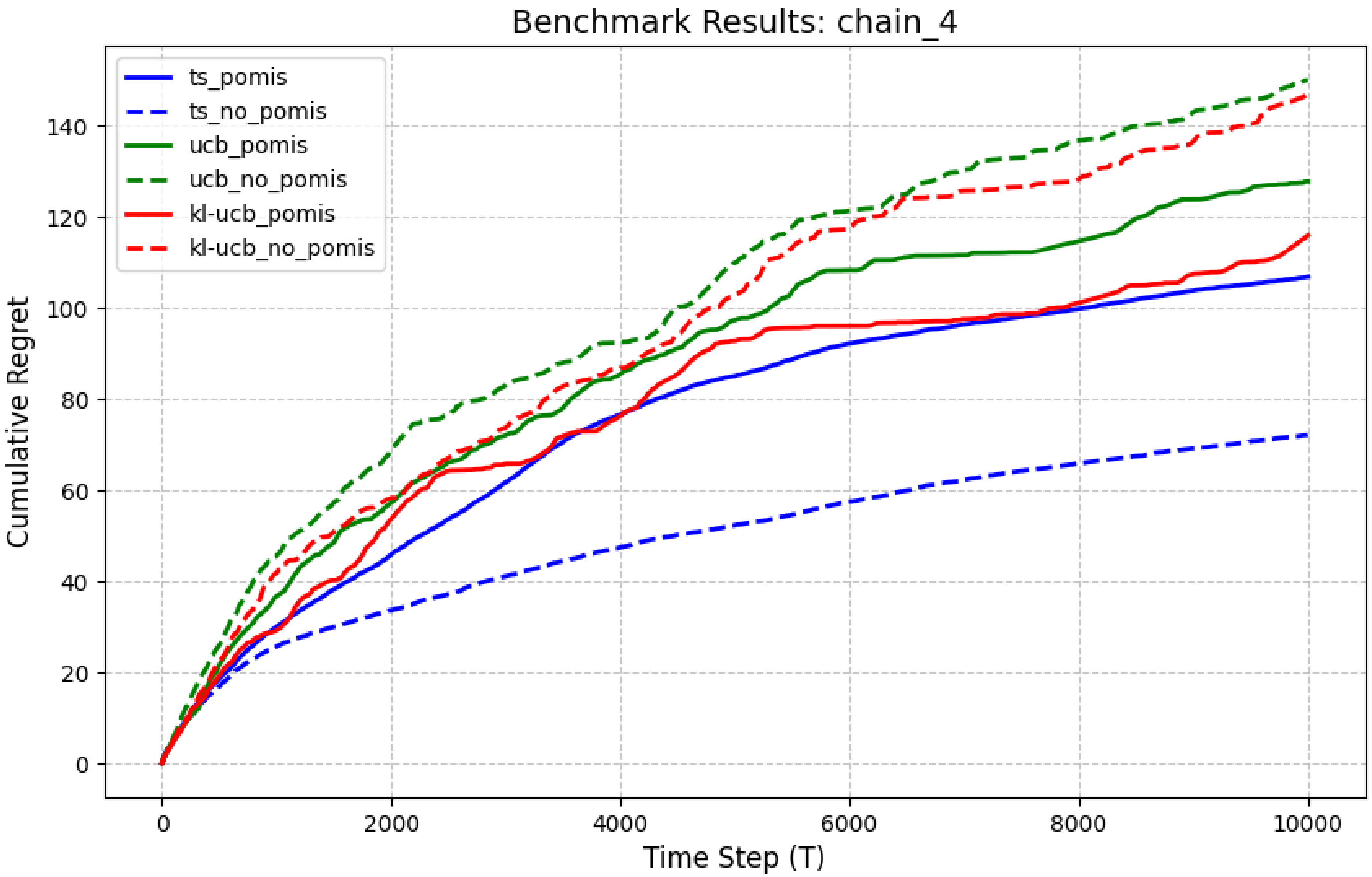
IV Case

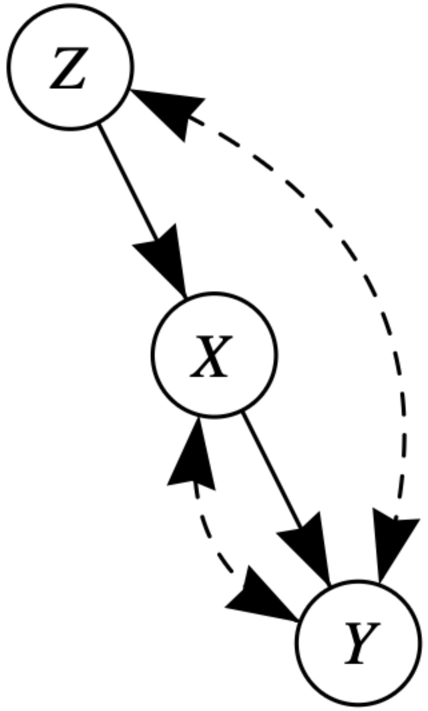




Experiments with UCB, KL-UCB, TS

Anomaly : Doubly Confounded Case





Anomaly Analysis

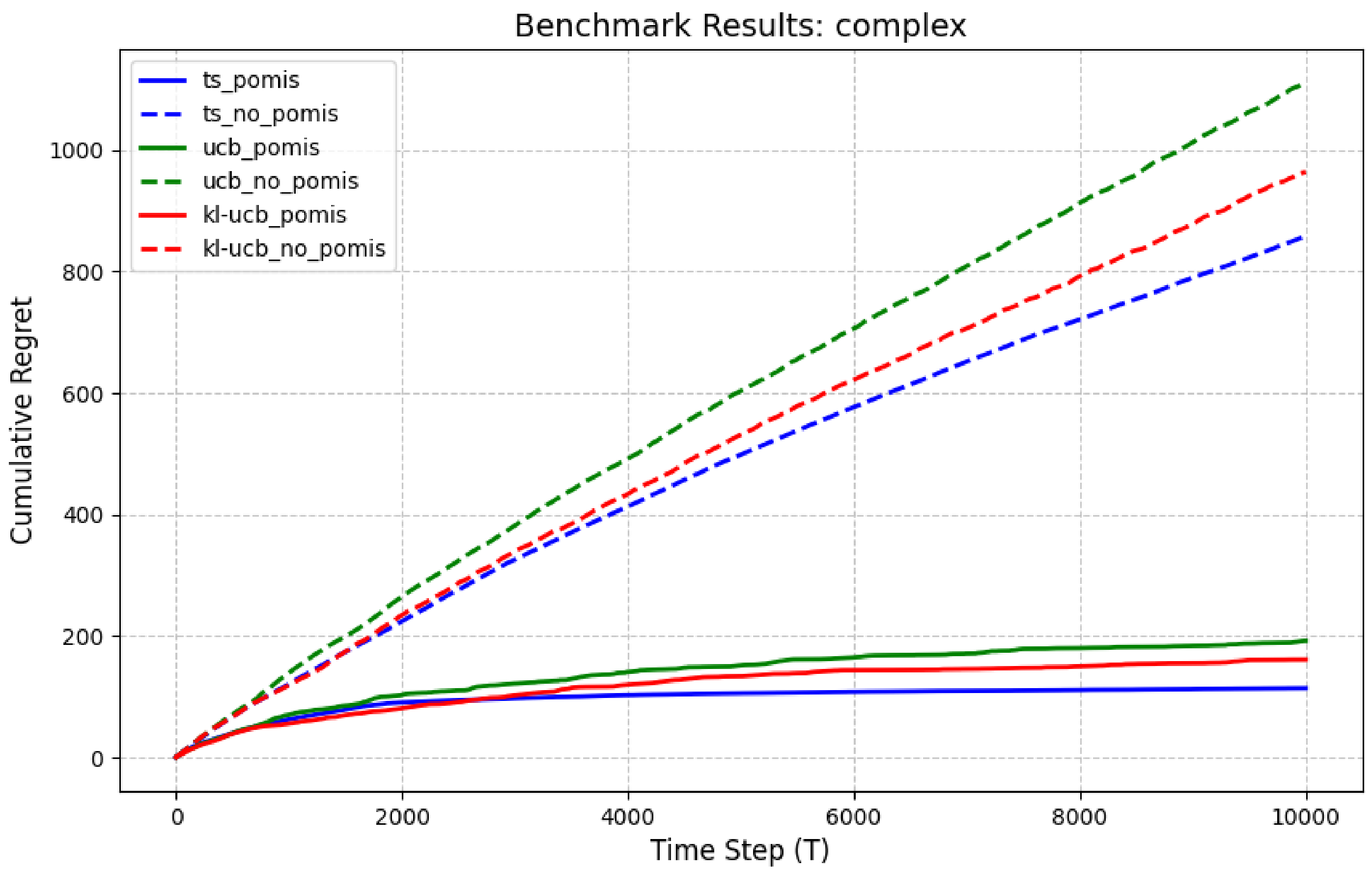
Thompson Sampling without POMIS Filtering

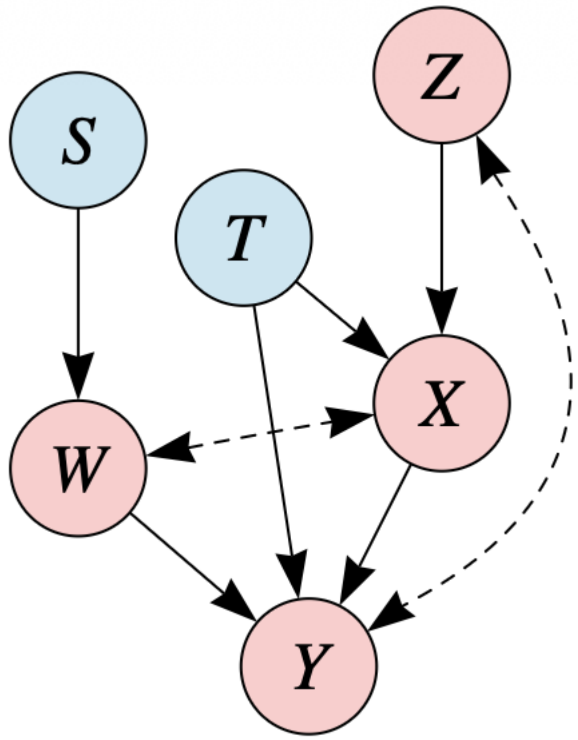
- Benefit from **Exploration Redundancy**
  - Multiple Optimal Arms are contained
  - In environment with high reward variance capturing one of optimal arms can be better strategy

- Risk Dispersion
  - POMISs reduces action space but only one optimal arm stands for it
  - Due to variance and bias, we might assume optimal arm suboptimal
  - However, without POMIS filtering, it works as an **Ensemble** so risk is dispersed

Experiments with UCB, KL-UCB, TS

Complex Case





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